

ELISAVET LYDIA ALVANAKI

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EDUCATION

Columbia University

New York, USA

PhD Student - Computer Science

September 2024 - June 2029(expected)

- Advisor: Prof. Luca Carloni

GPA: 4.04/4

National Technical University of Athens

Athens, Greece

Diploma - Electrical and Computer Engineering

September 2018 - 2023

- Advisors: Prof. Dimitrios Soudris, Prof. Sotirios Xydis
- Thesis title: *Decoupled Access-Execute and Dynamic Voltage/Frequency Scaling Optimization for Energy Efficient tinyML Deployments on STM32 MCUs*

GPA: 8.38/10, Major GPA: 9.05/10

PUBLICATIONS

SLDB: An End-To-End Heterogeneous System-on-Chip Benchmark Suite for LLM-Aided Design

Elisavet-Lydia Alvanaki, Kevin Lee, Luca P. Carloni

Accepted as full paper at the IEEE International Conference on LLM-Aided Design (ICLAD '25).

Acceptance Rate: 34%

Decoupled Access-Execute enabled DVFS for tinyML deployments on STM32 microcontrollers

Elisavet-Lydia Alvanaki, Manolis Katsaragakis, Dimosthenis Masouros, Sotirios Xydis, Dimitrios Soudris

Accepted as full paper at Design Automation and Test in Europe (DATE '24). Acceptance Rate: 25%

RESEARCH EXPERIENCE

Graduate Research Assistant

New York, NY

System-Level Design Group - Columbia University

September 2024-May 2029(expected)

- Research interests: AI for EDA, Hardware/Software co-design, Computer Architecture, SoC design
- Developed SLDB, an end-to-end System-on-Chip benchmark suite to evaluate LLMs on system-level integration and configuration tasks.
- Conducted research on improving correctness of LLM-generated RTL for component-level and system-level design, leveraging ML techniques, formal verification and AI agents
- Conducted research on contrastive learning for RTL waveform debugging

Research Intern

Durham, NC (Remotely)

Cyber Physical Systems Lab - Duke University

September 2023-December 2023

- Advisor: Professor Miroslav Pajic
- Research internship on energy-efficient neuromorphic controllers with tunable spiking representations.

Research Assistant

Athens, Greece

Microprocessor Systems Laboratory - National Technical University of Athens

July 2022-November 2023

- Thesis research on energy-efficient edge AI for ARM Cortex-M microcontrollers
- Designed and implemented a DVFS framework for Arm Cortex-M, with optimized clock parameter selection
- Implemented a decoupled access-execute framework for DVFS on CNNs, integrated with an optimizer to select optimal DVFS strategies under distinct QoS constraints
- Achieved a 25.2% reduction in energy consumption compared to state-of-the-art inference engines for tinyML

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern - System Optimization

Tesla

Athens, Greece

March 2022-July 2022

- Developed software for multiobjective powertrain optimization using evolutionary algorithms, achieving an 18% motor torque improvement with the same number of epochs
- Benchmarked and developed software for various evolutionary optimization algorithms (SPEA-2, Differential Evolution etc.)

VOLUNTEERING EXPERIENCE

Embedded Systems Engineer

Prometheus Eco Racing NTUA Team

Athens, Greece

January 2021-March 2022

- Designed PCBs for the vehicle's electric circuits (inverter, smart wheel, CAN Bus and telemetry) using Altium
- Designed and developed software for motor predictive maintenance using embedded AI

TEACHING EXPERIENCE

Project Mentor

COMS6901-Projects in Computer Science

Columbia University

September 2025-Present

- Mentored a project on agentic data generation for LLM-aided waveform debugging.

Teaching Assistant

System-on-Chip Platforms

Columbia University

September 2025-Present

- Designed programming exercises for hardware acceleration of transformer workloads. Graded homework assignments related to Latency-Insensitive Design.

Teaching Assistant

Embedded Systems Design

Microprocessor Systems Lab - NTUA

September 2023-January 2024

- Designed lab exercises on design space exploration of code optimization parameters (tiling, loop unrolling etc.)

Lab Assistant

Microprocessor Systems Lab

Microprocessor Systems Lab - NTUA

September 2022-September 2023

- Assisted students with debugging AVR boards and conducted lab exams

HONORS & AWARDS

Mindspace Challenge Finalist

Mindspace Challenge

June 2019

- Created a seed stage startup (StudyBuddies), aimed at making peer-to-peer tutoring accessible to university students and won 3rd place among more than 100 teams

RELATED COURSEWORK

System-on-Chip Platforms, Embedded Scalable Platforms, Computer Vision, Computational Aspects of Robotics, Embedded Systems, Advanced Computer Architecture, Digital VLSI Systems, Microprocessor Systems, Parallel Processing Systems, Microprocessor Systems Lab, Distributed Systems, Artificial Intelligence

TECHNICAL SKILLS

Programming skills:

C, C++, Verilog, VHDL, Python, bash, Matlab

Other skills:

Unix/Linux, Tensorflow, TensorflowLite Micro, CMSIS-NN, Git, MCU Programming, Embedded Systems